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- In a Hidden Markov Model (HMM), the emission probabilities depend only on the current hidden state. (True/False)
- The Viterbi algorithm is used to compute the total probability of observing a sequence in an HMM. (True/False)
- The Forward algorithm can be used to find the single most probable hidden path for a given sequence. (True/False)
- In an HMM, which of the following defines a model completely?
 (A) Transition probabilities only (B) Emission probabilities only
 (C) Both transition and emission probabilities, plus initial state probabilities (D) Hidden states only
- The Forward–Backward algorithm in a Hidden Markov Model estimates the most probable hidden path by combining the forward and backward probabilities for each state at every position. (True/-False)
- If an HMM has 3 hidden states and the observed sequence has length n , the time complexity of Viterbi is:
 (A) $O(n)$ (B) $O(n \times 3^2)$ (C) $O(3^n)$ (D) $O(n^2)$
- Which problem does the Forward algorithm solve?
 (A) Finding the most probable hidden path
 (B) Computing total probability of the observed sequence
 (C) Parameter estimation
 (D) Identifying states with maximum emission
- Baum–Welch algorithm is an example of:
 (A) Supervised learning (B) Unsupervised learning (expectation–maximization)
 (C) Reinforcement learning (D) Gradient descent
- Which of the following steps is part of Viterbi decoding?
 (A) Summing probabilities over all paths
 (B) Taking maximum over incoming transitions
 (C) Estimating emission probabilities
 (D) Using pseudocounts to smooth transitions
- Soft decoding in HMMs refers to:
 (A) Finding the most probable hidden path
 (B) Computing the probability that each state was visited at each position
 (C) Estimating emissions only
 (D) Using pseudocounts to regularize parameters